A DANGERous Data Challenge

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DANGER: Data, Numbers, and Geometry, BIRS
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AI for math

Artificial intelligence and machine learning tools are used in a variety of different ways to help us gain insight and understanding, generate conjectures, and prove theorems.

- Analyze data
  - Search for patterns and correlations (data science)
  - Classification (model training)
- Construct examples
  - Generate extremal/interesting examples (Funsearch, PatternBoost, AlphaEvolve, ...)
  - Extend known or suspected patterns, test conjectures
- Prove theorems
  - Informally (First Proof solutions by OpenAI and DeepMind/Aletheia, Bryan-Elek-Manners-Salafatinos-Vakil, Swaminathan, . . .)
  - Formally (AlphaProof, Aristotle, Ax-Prover, AxiomProver, DeepSeek-Prover, Gauss, Gödel-Prover, Kimina-Prover, Seed-Prover, . . .)

It's all 0s and 1s

Mathematics has two key features that make it especially attractive for AI:

- An inexhaustible supply of noiseless data.
- Computational access to ground truth.

What is data?

In mathematics, data are not observations, they are facts (aka theorems). That the elliptic curve $y^2 + xy + y = x^3 + x^2 - 10x - 10$ has rank 0 is one of 3,824,372 similar facts one can find in the LMFDB. Like all mathematical facts, it can be encoded as a sequence in some fixed alphabet (WLOG, we can assume that alphabet is $\{0, 1\}$).

What is a proof?

A proof (at least in modern mathematics) is a series of permissible steps that allow one to derive a stated conclusion from a stated hypothesis and an agreed list of axioms. Every proof can (at least in principle) be encoded as a sequence in some fixed alphabet.

Warmup example: compositeness

Fact: 91 is composite.
Proof: $91 = 7 \times 13$.
Better proof: $2^{91} \not\equiv 2 \bmod 91$
The integer 2 witnesses/certifies that 91 is composite.[1]

The triple $(91, 7, 13)$ encodes a proof of the compositeness of 91 that can be mechanically translated into a formal proof with a quasi-linear expansion factor.

The pair $(91, 2)$ also encodes a proof of the compositeness of 91 that can be mechanically translated into a formal proof, with a quasi-quadratic expansion factor.

The singleton (91) can also be viewed as encoding a proof of the compositeness of 91 that can be mechanically translated into a formal proof, but with an exponential expansion factor: $2|91 \vee 3|91 \vee 4|91 \vee 5|91 \vee 6|91 \vee 7|91 \vee 8|91 \vee 9|91$.

[1] Not all composite numbers have a witness of this type, in general one uses the more general notion of a Rabin-Miller witness.

Primality proving

Fact: 89 is prime.
Non-proof: $89 = 1 \times 89$
Non-proof: $2^{89} \equiv 2 \bmod 89$
Proof: $2^{88} \equiv 1 \bmod 89 \ \wedge \ 2^{44} \not\equiv 1 \bmod 89 \ \wedge \ 2^8 \not\equiv 1 \bmod 89$
$\wedge\ 89 - 1 = 88 = 2^3 \times 11 \wedge 2$ is prime $\wedge\ 11$ is prime
Better proof: The point $P := (6, 45)$ on the elliptic curve $y^2 = x^3 + 13x^2 + x$ satisfies $2^4 P \neq 0$ and $2^5 P = 0$, with $2^5 > q + 1 + \lfloor 2\sqrt{q} \rfloor$ where $q = \lfloor \sqrt{p} \rfloor = 9$.

The Pomerance triple $(89, 13, 6)$ encodes a proof of the primality of 89 that can be mechanically translated into a formal proof, with a quasi-quadratic expansion factor.

Pomerance proofs are the **gold standard** of primality proofs; every other type (Pratt, ECPP, AKS, ...) is more complicated and has a larger expansion factor.

Pomerance proofs

For a positive integer p, define $k_p := \lceil \log_2(q + 1 + 2\sqrt{q}) \rceil$ where $q = \lfloor \sqrt{p} \rfloor$.

Definition

A triple of integers (p, A, x_0) with $x_0, A \in [0, p-1]$ and $A^2 \not\equiv \pm 2 \bmod p$ is a Pomerance triple if there exist integers $B, y_0 \in [0, p-1]$ such that $P := (x_0, y_0)$ is a solution to $E : By^2 = x^3 + Ax^2 + x$, with $2^{k_p} P = 0$ and $2^{k_p - 1} P \neq 0$.[2]

Theorem

If (p, A, x_0) is a Pomerance triple with $p > 2$ odd, then p is prime.

Proof.

$A \not\equiv \pm 2 \bmod p$ ensures E is non-singular modulo any odd divisor q of p.
If p is not prime then we may choose $q \leq \sqrt{p}$. The point $P = (x_0, y_0)$ has order 2^{k_p} which is larger than $\#E(\mathbb{F}_q) \leq q + 1 + 2\sqrt{q}$, a contradiction. □

[2] The inequality $2^{k_p - 1} P \neq 0$ must hold over $\mathbb{Z}/q\mathbb{Z}$ for every $q | p$.

Pomerance proofs

Note that $By_0^2 = x_0^3 + Ax_0^2 + x_0$ implies that the square class of B in $(\mathbb{Z}/p\mathbb{Z})^\times$ is determined by A and x_0, and this uniquely determines the isomorphism class of $By^2 = x^3 + Ax^2 + x$, so we can take B to be any representative of its square class, and this determines y_0 up to sign.

But in fact we never need to know the values of B and y_0!

Given a projective point $P = (x_1 : y_1 : z_1)$ on $By^2z = x^3 + Ax^2z + xz^2$, we can compute $2P = (x_2 : y_2 : z_2)$ by letting $C := (A + 2)/4$ and applying

$$\begin{aligned} u &= (x_1 + z_1)^2 \\ v &= (x_1 - z_1)^2 \\ w &= u - v \\ x_2 &= uv \\ z_2 &= w(v + Cw) \end{aligned}$$

Certifying a Pomerance triple

```python
def pp_verify(p, A, x0):
    from math import gcd, isqrt

    if p < 5 or p % 2 == 0:
        return False

    q = isqrt(p)                                # p not prime ==> prime divisor < q
    k = (q + 1 + isqrt(4*q)).bit_length()   # least k such that 2^k > q + 1 + 2*sqrt(q)

    if gcd(A * A - 4, p) != 1:
        return False                            # singular

    X, Z = x0 % p, 1
    Zprev = None

    # precompute C = (A + 2)/4 mod p for doubling formula
    C = ((A + 2) * ((p + 1) // 4 if p % 4 == 3 else (3*p +1) //4)) % p
    for i in range(k):
        Zprev = Z
        U,V = (X+Z)*(X+Z) % p, (X-Z)*(X-Z) % p
        W = U - V
        X,Z = U*V % p, W*(V+C*W) % p

    # if p|Z and (Zprev,p)=1 then (p,A,x0) is valid
    return Z % p == 0 and gcd(Zprev, p) == 1
```

Demonstration

Code and data are available in the following GitHub repo:

https://github.com/AndrewVSutherland/DANGER3

```
python3 vpp.py 89 13 6
python3 lean_vpp.py 89 13 6 >pp89.lean
```

Online Lean4/Mathlib calculator available at:

https://live.lean-lang.org/

The landscape of Pomerance triples

Theorem
Pomerance triples (p, A, x_0) *exist for every prime* $p > 3$.

For any fixed p, there are on the order of $\sqrt{p}$ integers $A \in [0, p-1]$ that appear in a Pomerance proof for p,[3] and for each such A there are $2^{k_p - 1} \approx \sqrt{p}$ possibilities for x_0. Thus every prime $p > 3$ appears in roughly p distinct Pomerance triples.

Given a prefix (p, A) of a Pomerance triple, it is easy to find an x_0 for which (p, A, x_0) is a Pomerance triple (this takes quasi-quadratic expected time).

For a general prime p, no efficient algorithm is known for finding a Pomerance triple (p, A, x_0). The fastest known general algorithm simply picks A's at random and tries to extend (p, A) to (p, A, x_0) (this will fail or succeed in quasi-quadratic expected time).

[3] "On the order of" ignores factors that are $O(\log p)$.

Data challenge

Challenge: Given a (potentially large) prime p, generate a Pomerance triple (p, A, x_0).

Resources available at https://github.com/AndrewVSutherland/DANGER3:

- vpp.py: Python program to test candidate triples.
- pp10.txt: the 236,264 Pomerance triples with $p < 2^{10}$.
- pp20.txt: one Pomerance triple for each of the 82,023 primes $3 < p < 2^{20}$.
- pp12.txt.gz: the 3,083,880 Pomerance triples with $p < 2^{12}$.
- pp24.txt.gz: one Pomerance triple for each of the 1,077,869 primes $3 < p < 2^{24}$.
- pp16A.txt.gz: the 4,713,069 prefixes (p, A) of Pomerance triples with $p < 2^{16}$.

Resources available elsewhere (links available in the GitHub repo):

- pp14.txt.gz: the 43,307,624 Pomerance triples with $p < 2^{14}$ (646MB).
- pp16.txt.gz: the 598,705,640 Pomerance triples with $p < 2^{16}$ (9887MB).
- pp28.txt.gz: one Pomerance triple for 14,630,841 primes $3 < p < 2^{28}$ (386MB).